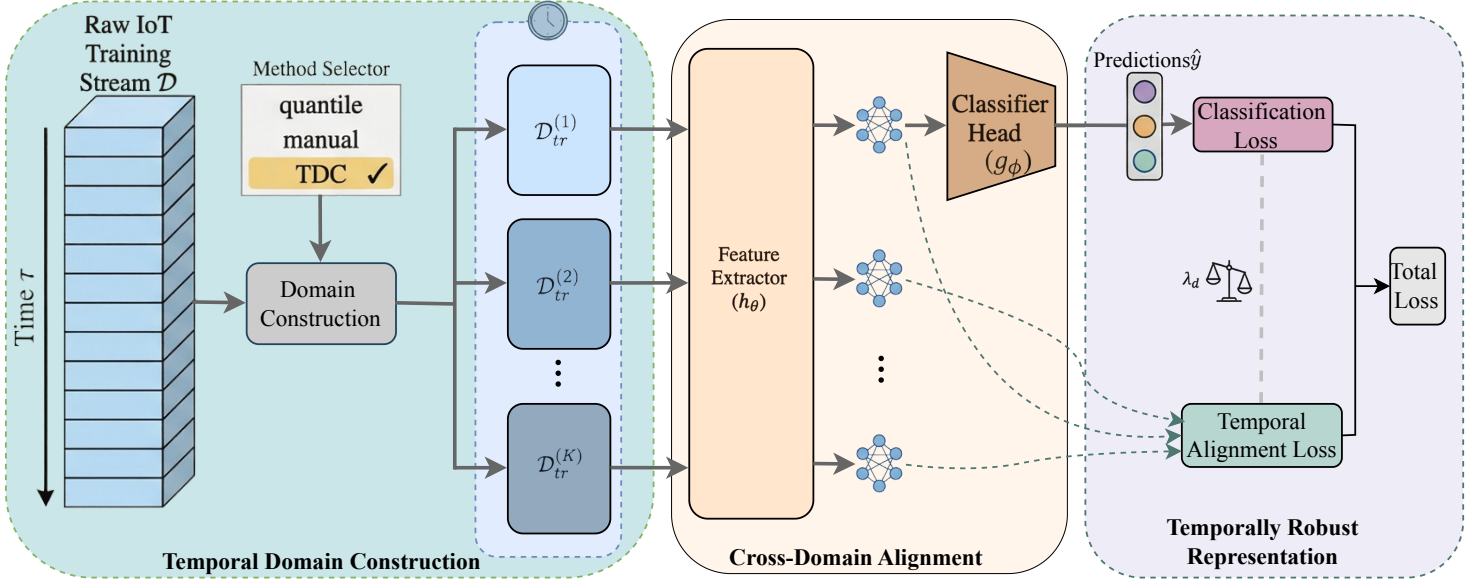
Raw IoT Training Stream $\mathcal{D}$
Time τ
Method Selector
quantile
manual
TDC ✓
Domain Construction
$\mathcal{D}_{tr}^{(1)}$
$\mathcal{D}_{tr}^{(2)}$
$\mathcal{D}_{tr}^{(K)}$
Temporal Domain Construction
Feature Extractor (h_θ)
Classifier Head (g_ϕ)
Cross-Domain Alignment
Predictions $\hat{y}$
Classification Loss
λ_d
Total Loss
Temporal Alignment Loss
Temporally Robust Representation